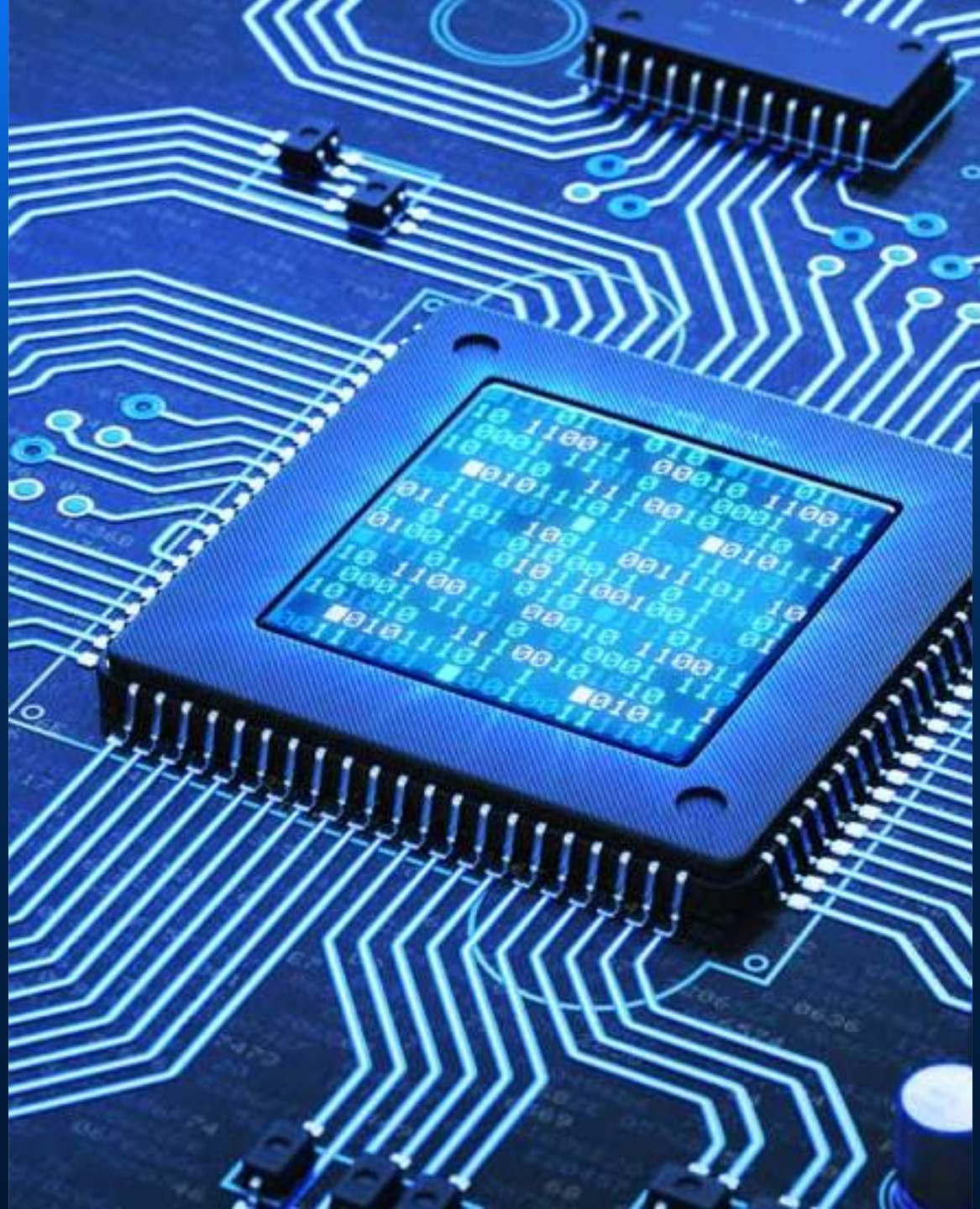


Computer Architecture

Session 9

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Senior HW Digital Design Engineer

Agenda

Timing Analysis

1 Understanding of STA concept

2. Applying STA for MIPS Example

Agenda

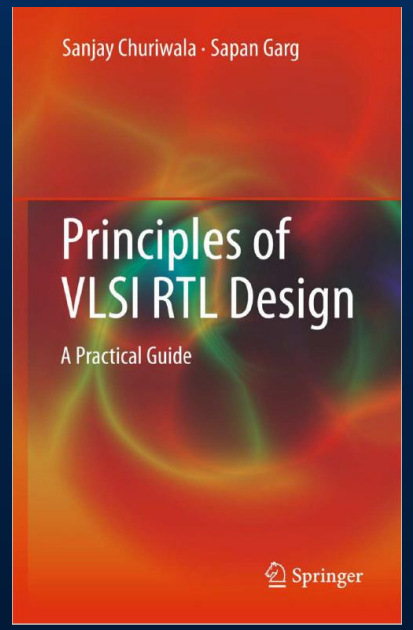
Timing Analysis

1 Understanding of STA concept

2. Applying STA for MIPS Example

This presentation is prepared from:

Principles of VLSI RTL Design.



Agenda

3. Timing Analysis

- # 3.1 Scope of STA
- # 3.2 Timing Parameters for Digital Logic
- # 3.4 Sequential Arcs
- # 3.5 Understanding Setup and Hold
- # 3.7 Basic Analysis
- # 3.8 Uncertainty

3.1 Scope of STA

3.1 Scope of STA

- The scope of STA is limited to only validate the ability to meet the desired timing goals.
- STA does not do anything to deal with the logical functionality of the design.
- Assume that the following RTL code excerpt is taken through RTL simulation.
- The simulation will simply assume that during the next cycle, a and b will reach the register bank S.

```
always @(posedge clock)
S <= a * b;
```

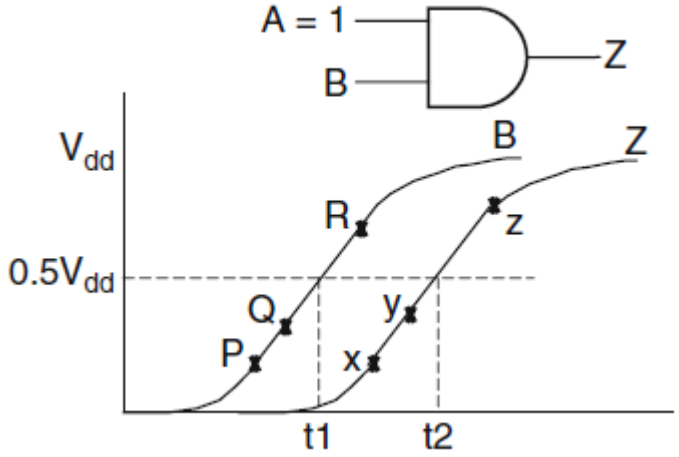
- The simulator will not validate this assumption, even if the clock frequency is increased drastically, or, even if the width of a and b are increased, or even if there is more complex logic in the data-path.
- So, after the RTL simulation has passed, you do not yet know – whether the design can exhibit the same desired functionality at the required frequency.

3.2 Timing Parameters for Digital Logic

3.2 Timing Parameters for Digital Logic

Delay Parameters for Digital Logic

- Delay is defined as the time between the application of an input and the observation of the response at the output.
- There is no specific point which has to be chosen as the reference.
- For CMOS designs, this happens during the middle part of the transitions.
- Usually, these points are expressed as a percentage or fraction of Vdd.
- In this example, 0.5Vdd has been chosen as the point of mea well as t2.
- It is not necessary that both t1 and t2 should have the same fraction.
- One could decide to use 0.5Vdd for (t1) and 0.4Vdd for (t2).



3.2 Timing Parameters for Digital Logic

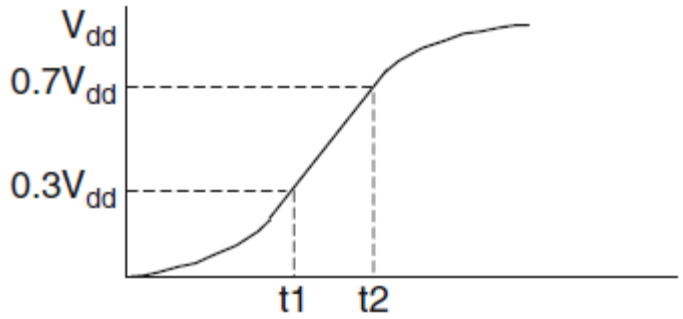
Delay Parameters for Digital Logic

- The excerpt below shows an example of the delay measurement point specification in a hypothetical timing library (in SLF format)
  - input_threshold_pct_fall : 45 ;
  - output_threshold_pct_fall : 45 ;
  - input_threshold_pct_rise : 45 ;
  - output_threshold_pct_rise : 45 ;

3.2 Timing Parameters for Digital Logic

Slew Parameters for Digital Logic

- Slew (or transition time or ramp time) refers to the time that it will take for the signal to go from 0 to 1 (and vice-versa).



- There are no standard measurement points.
- The slew specification should include the points at which measurement was started and at which the measurement was stopped.
- In this example, t1 was measured at 0.3Vdd and t2 was measured at 0.7Vdd.
- Some of the more commonly used combinations are: 0.1Vdd – 0.9Vdd; 0.2Vdd – 0.8Vdd: 0.3Vdd – 0.5Vdd; 0.3Vdd – 0.7Vdd.

3.2 Timing Parameters for Digital Logic

Slew Parameters for Digital Logic

- The excerpt below shows an example of the slew measurement point specification in a hypothetical timing library:
  - slew_lower_threshold_pct_rise : 35 ;
  - slew_lower_threshold_pct_fall : 35 ;
  - slew_upper_threshold_pct_rise : 65 ;
  - slew_upper_threshold_pct_fall : 65 ;
- So, if you are comparing the speed of two different libraries or two different data sheets, don’t just look at the delay or slew numbers. Also, look at the points at which the measurements were made.

3.4 Sequential Arcs

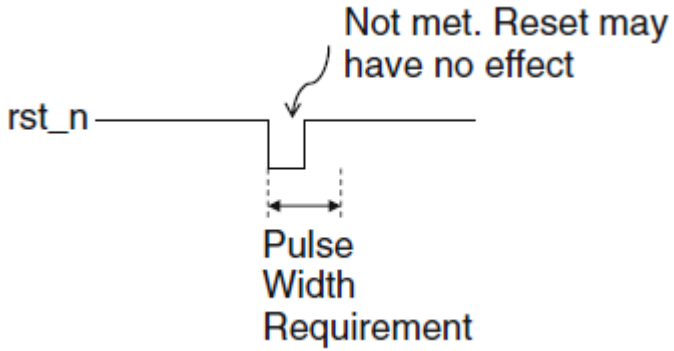
3.4 Sequential Arcs

- There are timing relationships between two inputs themselves or multiple edges on the same input.
- These relationships usually exist only for sequential cells.
- These are various kinds of requirements (timing checks) that need to be satisfied so that the sequential cell behaves reliably.
- These checks are specified in tables whose entries are based on input slews of the signals – between which the timing check has to be made.
- These arcs are:
  - Pulse Width.
  - Setup
  - Hold
  - Recovery
  - Removal

3.4 Sequential Arcs

Pulse Width

- It's the minimum width of the pulse on clocks and asynchronous pins, so that the sequential cell can register the clock or an assertion of the asynchronous pin.
- Example: assume an active Low asynchronous reset There would be a requirement of minimum width for a Low Pulse.

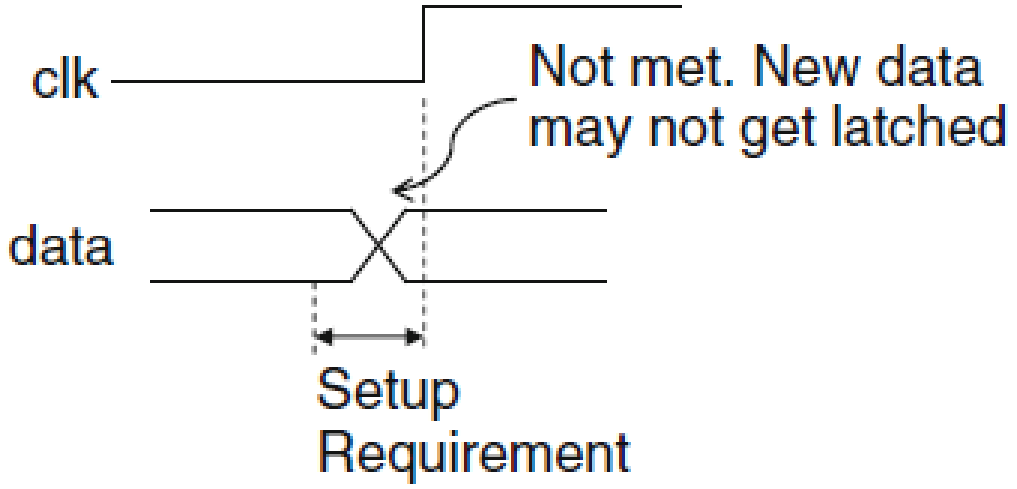


- If a pulse is applied which is of duration shorter than the minimum Pulse Width, the output may not go to the desired level.

3.4 Sequential Arcs

Setup

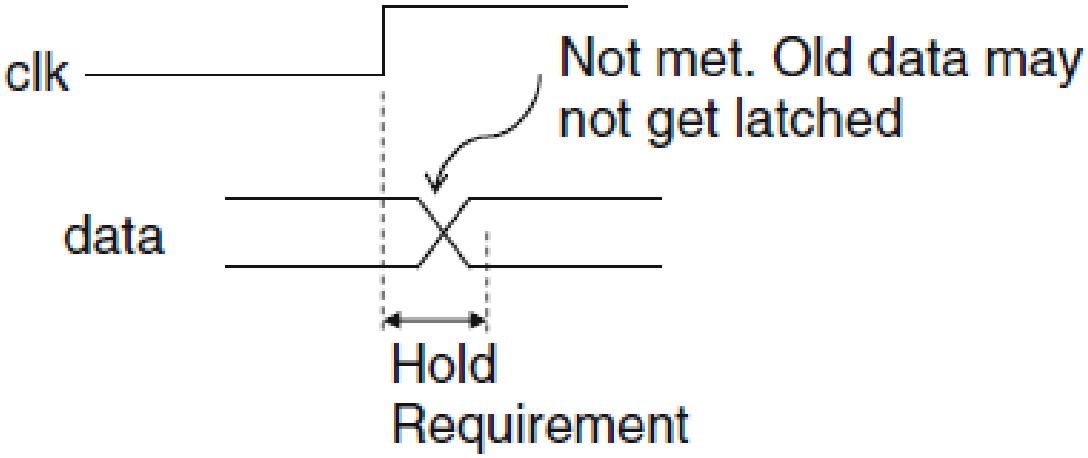
- It’s the minimum duration before the triggering edge of clock, by which the data should have come and stabilized.
- If the data changes within this duration, there is no guarantee that the new data would get captured in the sequential device.



3.4 Sequential Arcs

Hold

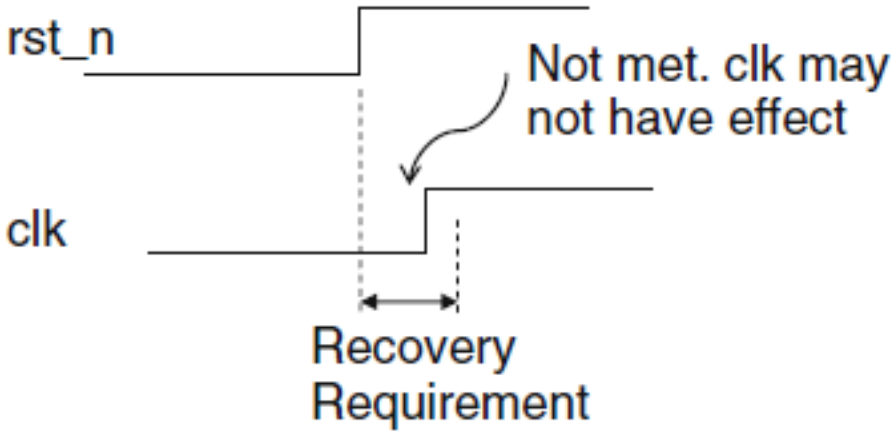
- It’s the minimum duration after the triggering edge of clock, till which the data should retain its stable value.
- If the data changes within this duration, there is no guarantee that the new data would not interfere with the data being captured within the sequential device.



3.4 Sequential Arcs

Recovery

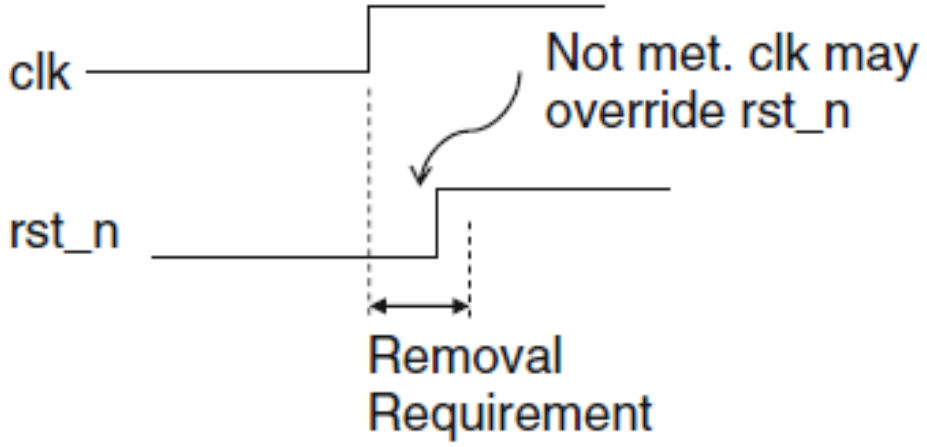
- It’s the minimum duration after the de-assertion of an asynchronous control signal, during which the clock’s triggering edge should not come.
- Looked alternately, it also means, the time before the clock triggering edge, by when asynchronous control pin should have been de-asserted.
- Hence, this might be considered as a setup check.



3.4 Sequential Arcs

Removal

- It’s the minimum time after a clock’s triggering edge – for which an asynchronous control pin should remain asserted.
- As can be seen, this might also be thought of as a hold check.

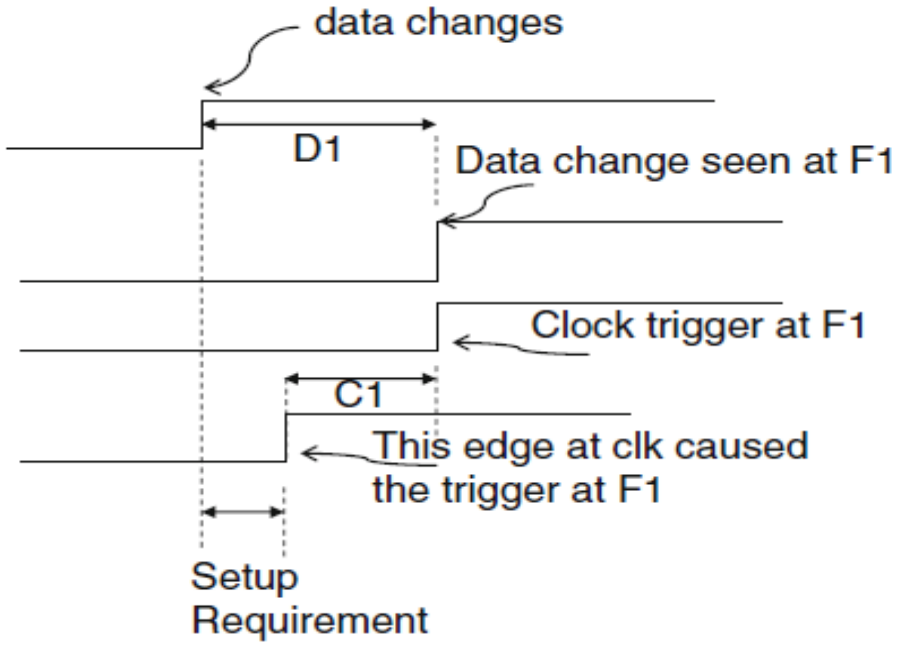


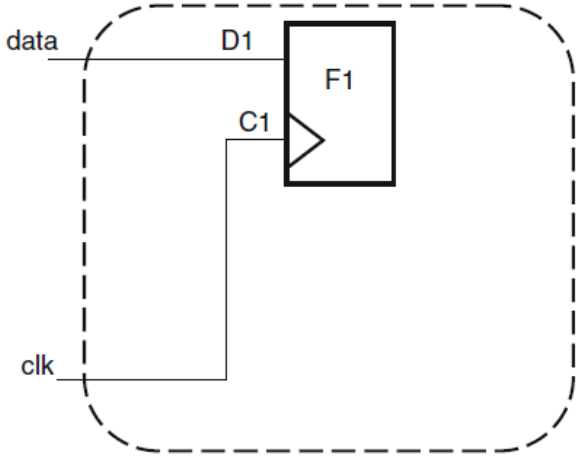
3.5 Understanding Setup and Hold

3.5 Understanding Setup and Hold

Understanding Setup

- First assume C1=0, clk will reach the flop's clock terminal instantaneously. However, data has still not reached the corresponding data terminal.
- data should be available at time D1 before the triggering edge appears at clk.
- Setup = Max(all data path delays) – Min(all clock path delays)

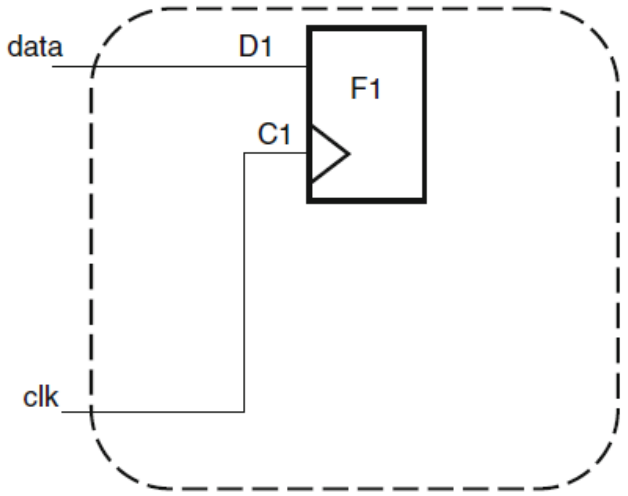




3.5 Understanding Setup and Hold

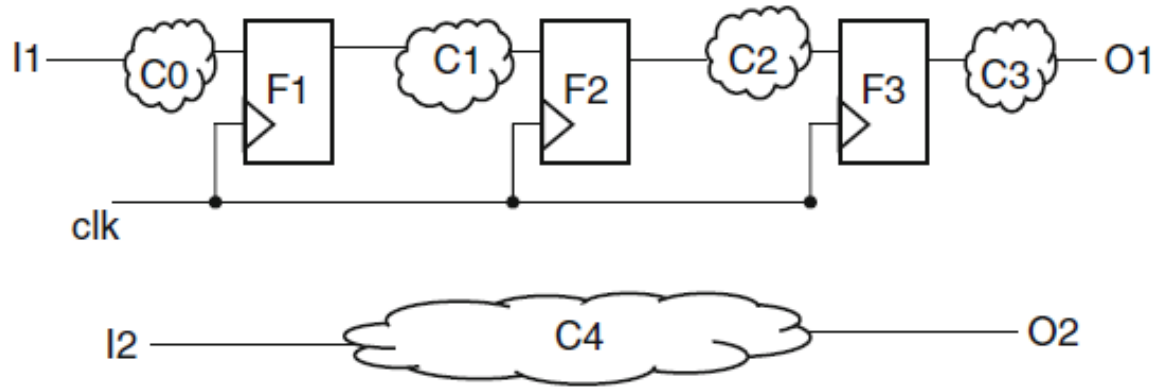
Understanding Hold

- First assume D1=0, new data will reach the flop's data terminal instant. However, clk still not reached the corresponding clock terminal.
- old data should be available at time D1 after the triggering edge at clk.
- Hold = Max(all clock path delays) – Min(all data path delays)



3.7 Basic Analysis

3.7 Basic Analysis



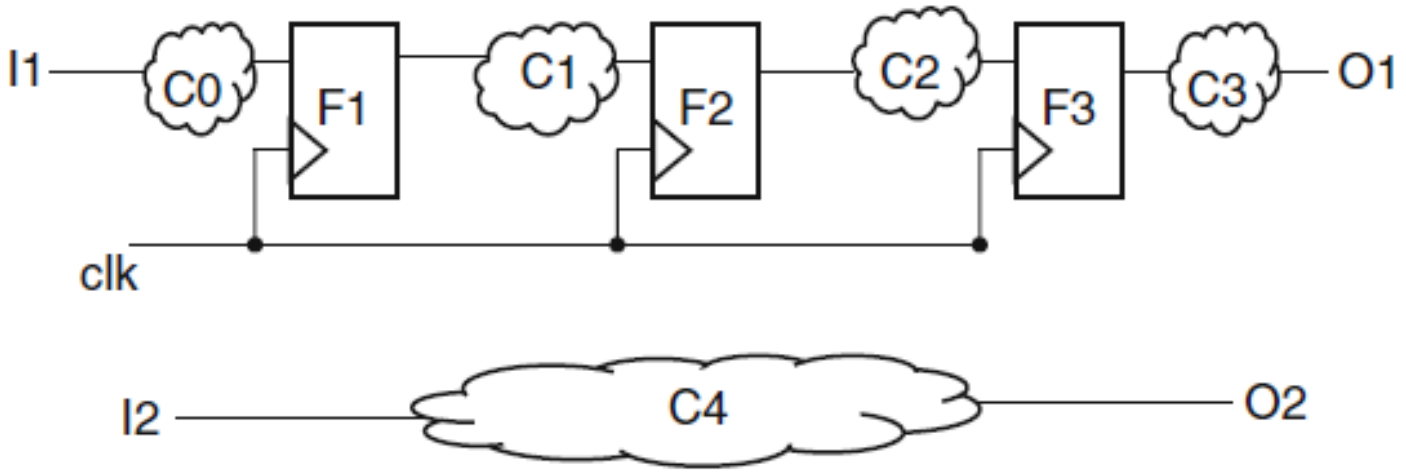
For F1 to capture data reliably:

- T_I1 + C0 ≤ P - S1 // so that the data is available by the time next clock edge
- T_I1 + C0 ≥ H1 // so that a new data does not corrupt the previous value

This is also expressed as:

- Setup slack = P - S1 – T_I1 – C0
  A positive setup slack value means the setup requirement has been met.
- Hold slack = T_I1 + C0 – H1
  A positive hold slack value means the hold requirement has been met.

3.7 Basic Analysis



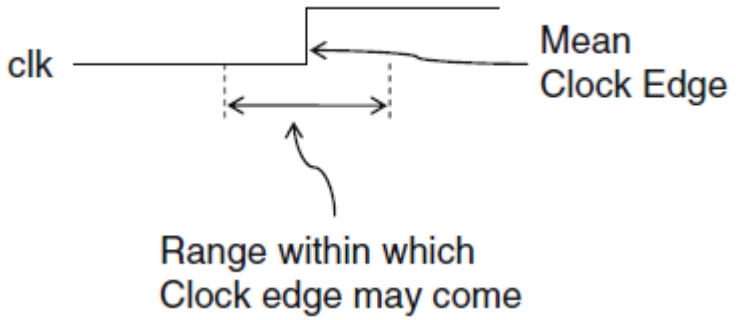
For F2 to capture data reliably:

- D_Q1 + C1 ≤ P - S2 and
- D_Q1 + C1 ≥ H2
- Setup slack = P - S2 – D_Q1 – C1
- Hold slack = D_Q1 + C1 – H2

3.8 Uncertainty

3.8 Uncertainty

- Uncertainty refers to the variation in the actual triggering edge of the clock.
- Uncertainty occurs due to:
  - PLL Jitter
  - Skew in different clock paths.



- PLL Jitter is a slight fluctuation in the time at which the edges occur due to various physical parameters. So, the time between two consecutive active edges of a clock may vary around the mean Period (*Period* ± *Jitter*).
- Skew defines the differential delay on the clock paths for the flops.

3.8 Uncertainty

For a setup analysis:

- In the worst-case scenario, the time between consecutive active edges of clock would be reduced by the maximum range of variation (**Jitter** + **Skew**).
- You should specify the combined impact of both jitter and skew as uncertainty.

For hold analysis:

- The same edge of clock is used for both launch as well as capture.
- The uncertainty is only due to **skew**.

For both setup and hold analysis:

- The slack will be reduced by *the specified uncertainty*, which is smaller during hold analysis compared to its used value for setup analysis.

Agenda

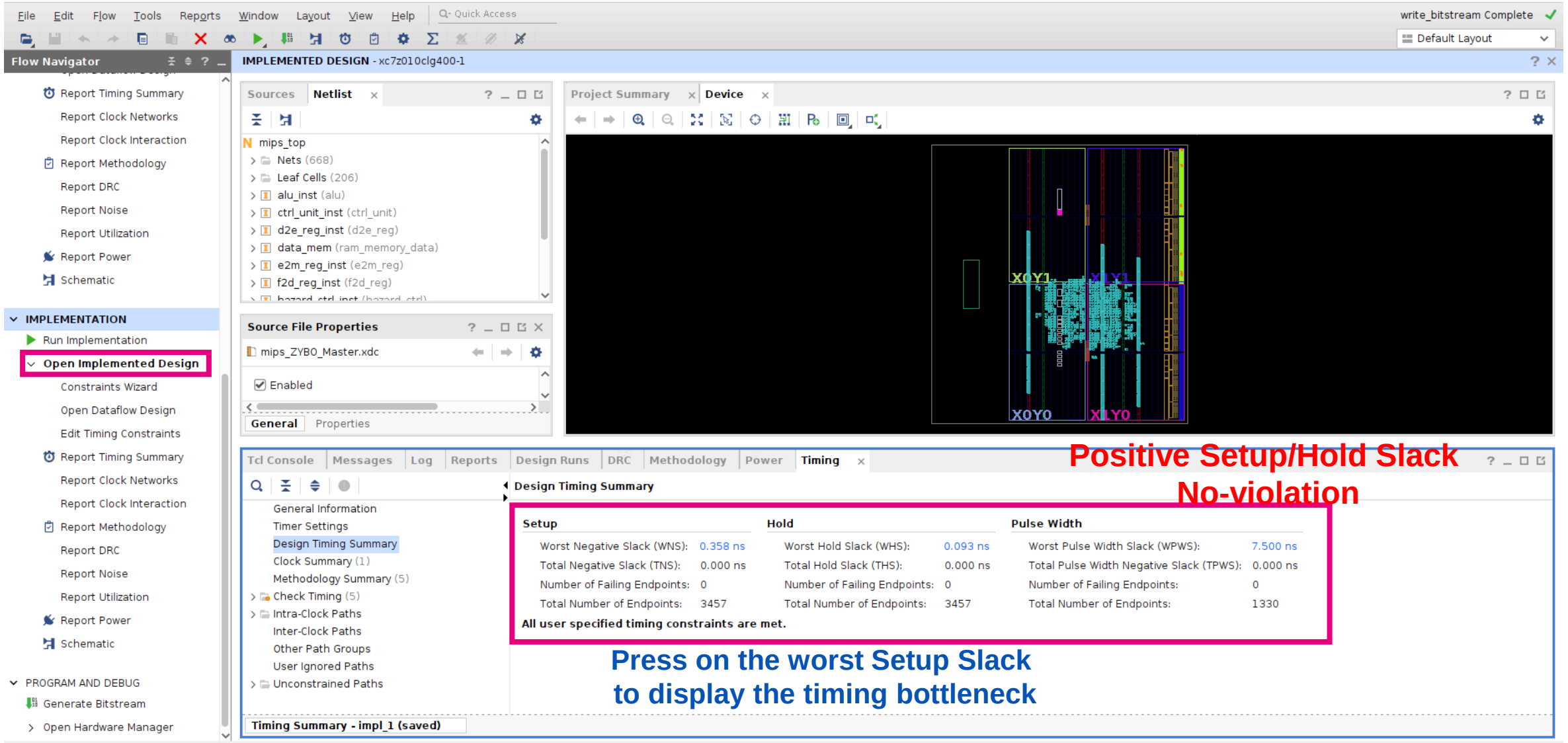
Timing Analysis

1 Understanding of STA concept

2. Applying STA for MIPS Example

Applying STA for MIPS Example

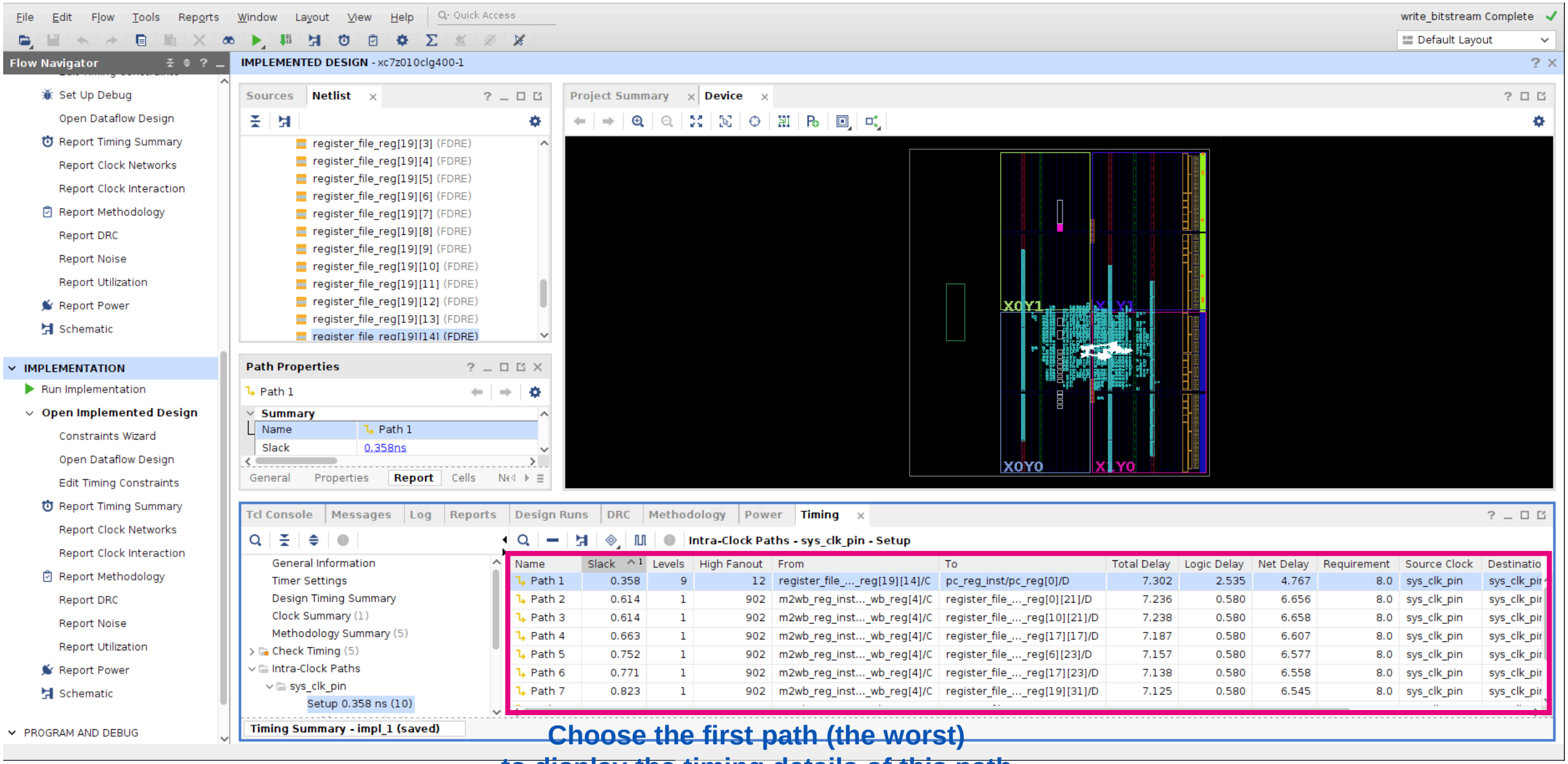
Applying STA for MIPS Example



Positive Setup/Hold Slack
No-violation

Press on the worst Setup Slack
to display the timing bottleneck

Applying STA for MIPS Example



Choose the first path (the worst)
to display the timing details of this path

Applying STA for MIPS Example

Project Summary | Device | **Path 1 - mips_top_timing_summary_routed**

Summary

Name	Path 1
Slack	0.358ns
Source	register_file_inst/register_file_reg[19][14]/C (falling edge-triggered cell FDRE clocked by sys_clk_pin {rise@0.000ns fall@8.000ns period=16.000ns})
Destination	pc_reg_inst/pc_reg[0]/D (rising edge-triggered cell FDCE clocked by sys_clk_pin {rise@0.000ns fall@8.000ns period=16.000ns})
Path Group	sys_clk_pin
Path Type	Setup (Max at Slow Process Corner)
Requirement	8.000ns (sys_clk_pin rise@16.000ns - sys_clk_pin fall@8.000ns)
Data Path Delay	7.302ns (logic 2.535ns (34.715%) route 4.767ns (65.285%))
Logic Levels	9 (CARRY4=2 LUT4=1 LUT5=1 LUT6=4 MUXF7=1)
Clock Path Skew	-0.061ns
Clock Un...rtainty	0.035ns

Source Clock Path

Delay Type	Incr (ns)	Path (...	Location	Cell ...	Cell	Netlist Resources
(clock sys_clk_pin fall edge)	(f) 8.000	8.000				
	(f) 0.000	8.000	Site: L16	clk		clk
net (fo=0)	0.000	8.000				clk
IBUF (Prop_ibuf_I_O)	(f) 1.491	9.491	Site: L16	O	clk_IBUF_inst (IBUF)	clk_IBUF_inst/O
net (fo=1, routed)	2.076	11.567				clk_IBUF
BUFG (Prop_bufg_I_O)	(f) 0.101	11.668	Site: BU...RL_X0Y16	O	clk_IBUF_BUFG_inst (BUFG)	clk_IBUF_BUFG_inst/O
net (fo=1329, routed)	1.674	13.342				register_file_inst/clk

Choose the first path (the worst)
to display the timing details of this path

Applying STA for MIPS Example

Source Clock Path

Delay Type	Incr (ns)	Path (...	Location	Cell ...	Cell	Netlist Resources
(clock sys_clk_pin fall edge)	(f) 8.000	8.000				
	(f) 0.000	8.000	Site: L16	clk		clk
net (fo=0)	0.000	8.000				clk
IBUF (Prop_ibuf_I_O)	(f) 1.491	9.491	Site: L16	O	clk_IBUF_inst (IBUF)	clk_IBUF_inst/O
net (fo=1, routed)	2.076	11.567				clk_IBUF
BUFG (Prop_bufg_I_O)	(f) 0.101	11.668	Site: BU...RL_X0Y16	O	clk_IBUF_BUFG_inst (BUFG)	clk_IBUF_BUFG_inst/O
net (fo=1329, routed)	1.674	13.342				register_file_inst/clk
FDRE			Site: SLICE_X26Y39	C	register_file_reg[19][14] (FDRE)	register_file_inst/register_file_reg[19][14]/C

Data Path

Delay Type	Incr (ns)	Path (	Location	Cell Pin	Cell	Netlist Resources
FDRE (Prop_fdre_C_Q)	(r) 0.459	13.801	Site: SLICE_X26Y39	Q	register_file_reg[19][14] (FDRE)	register_file_inst/register_file_reg[19][14]/Q
net (fo=3, routed)	0.853	14.654				register_file_inst/register_file[19][14]
LUT6 (Prop_lut6_I0_O)	(r) 0.124	14.778	Site: SLICE_X27Y38	O	rd_data2[14]_INST_0_i_6 (LUT6)	register_file_inst/rd_data2[14]_INST_0_i_6/O
net (fo=1, routed)	0.000	14.778				register_file_inst/rd_data2[14]_INST_0_i_6_n_0
MUXF7 (Prop_muxf7_I0_O)	(r) 0.212	14.990	Site: SLICE_X27Y38	O	rd_data2[14]_INST_0_i_2 (MUXF7)	register_file_inst/rd_data2[14]_INST_0_i_2/O
net (fo=1, routed)	0.730	15.720				register_file_inst/rd_data2[14]_INST_0_i_2_n_0
LUT6 (Prop_lut6_I3_O)	(r) 0.299	16.019	Site: SLICE_X27Y39	O	rd_data2[14]_INST_0 (LUT6)	register_file_inst/rd_data2[14]_INST_0/O
net (fo=2, routed)	0.811	16.829				data_rd2_d[14]
LUT5 (Prop_lut5_I1_O)	(r) 0.124	16.953	Site: SLICE_X21Y38	O	pc_reg_inst_i_56 (LUT5)	pc_reg_inst_i_56/O
net (fo=1, routed)	0.548	17.501				pc_reg_inst_i_56_n_0
LUT6 (Prop_lut6_I5_O)	(r) 0.124	17.625	Site: SLICE_X21Y36	O	pc_reg_inst_i_46 (LUT6)	pc_reg_inst_i_46/O
net (fo=1, routed)	0.546	18.172				pc_reg_inst_i_46_n_0
LUT6 (Prop_lut6_I5_O)	(r) 0.124	18.296	Site: SLICE_X25Y36	O	pc_reg_inst_i_35 (LUT6)	pc_reg_inst_i_35/O
net (fo=1, routed)	0.000	18.296				pc_reg_inst_i_35_n_0
CARRY4 (Prop_c...4_S[0]_CO[3])	(r) 0.532	18.828	Site: SLICE_X25Y36	CO[3]	pc_reg_inst_i_20 (CARRY4)	pc_reg_inst_i_20/CO[3]
net (fo=1, routed)	0.000	18.828				pc_reg_inst_i_20_n_0
CARRY4 (Prop_carry4_CI_CO[2])	(r) 0.228	19.056	Site: SLICE_X25Y37	CO[2]	pc_reg_inst_i_14 (CARRY4)	pc_reg_inst_i_14/CO[2]
net (fo=12, routed)	0.813	19.868				equal_d
LUT4 (Prop_lut4_I3_O)	(r) 0.309	20.177	Site: SLICE_X29Y37	O	pc_reg_inst_i_12 (LUT4)	pc_reg_inst_i_12/O
net (fo=1, routed)	0.467	20.645				pc_reg_inst/pc_next[0]
FDCE			Site: SLICE_X31Y37	D	pc_reg[0] (FDCE)	pc_reg_inst/pc_reg[0]/D
Arrival Time		20.645				

Choose the first path (the worst) to display the timing details of this path

Applying STA for MIPS Example

- Consider the following:

Destination Clock Path

Delay Type	Incr (ns)	Path (...	Location	Cell ...	Cell	Netlist Resources
(clock sys_clk_pin rise edge)	(r) 16.000	16.000				
	(r) 0.000	16.000	Site: L16	clk		clk
net (fo=0)	0.000	16.000				clk
IBUF (Prop_ibuf_I_O)	(r) 1.421	17.421	Site: L16	O	clk_IBUF_inst (IBUF)	clk_IBUF_inst/O
net (fo=1, routed)	1.880	19.301				clk_IBUF
BUFG (Prop_bufg_I_O)	(r) 0.091	19.392	Site: BU...RL_X0Y16	O	clk_IBUF_BUFG_inst (BUFG)	clk_IBUF_BUFG_inst/O
net (fo=1329, routed)	1.498	20.890				pc_reg_inst/clk
FDCE			Site: SLICE_X31Y37	C	pc_reg[0] (FDCE)	pc_reg_inst/pc_reg[0]/C
clock pessimism	0.391	21.281				
clock uncertainty	-0.035	21.246				
FDCE (Setup_fdce_C_D)	-0.243	21.003	Site: SLICE_X31Y37		pc_reg[0] (FDCE)	pc_reg_inst/pc_reg[0]
Required Time		21.003				

Choose the first path (the worst)
to display the timing details of this path